

Research Interests

Chiplet Architecture/Interconnect, HW/SW Co-Design, Domain-Specific Accelerator, and ASIC Design

Education

New York University

PH.D STUDENT IN ELECTRICAL AND COMPUTER ENGINEERING

- Advisor: Prof. Austin Rovinski
- Lab: REALISE Lab

New York, USA

Aug 2026 - Present

Seoul National University of Science and Technology

M.S IN ELECTRONIC ENGINEERING

- Advisor: Prof. Seung Eun Lee
- Lab: Computer Architecture Laboratory
- Thesis: Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller

Seoul, Republic of Korea

Mar 2022 - Feb 2024

Seoul National University of Science and Technology

B.S IN ELECTRONIC ENGINEERING

- Advisor: Prof. Seung Eun Lee

Seoul, Republic of Korea

Mar 2015 - Feb 2022

Work Experience

MangoBoost

SOFTWARE ENGINEER

- Developed a Kubernetes-native storage orchestration platform (Go Operator, Helm, React dashboard) managing DAOS distributed storage on bare-metal clusters.
- Implemented node lifecycle automation including NVMe/SPDK device management, SCM memory sizing, and multi-engine hugepage allocation.
- Operated and deployed multi-node K8s clusters with end-to-end provisioning pipeline (kubespray, Docker, Helm).

Seoul, South Korea

Jan 2026 - July 2026

SYSTEM ARCHITECT

- Architected an NVMe/TCP initiator (NTI) with dynamic ARM-FPGA acceleration, allowing protocol processing to be flexibly offloaded depending on performance requirements and hardware availability. This ensured both efficiency and reliability in large-scale deployments (e.g., Ceph).
- Extended scalability from the original 2 subsystems to 18 FPGA-accelerated functions via SR-IOV, while achieving full 200G line-rate throughput in aggregated performance tests. On the ARM side, designed a flexible execution path supporting up to 130 concurrent functions to enable diverse application workloads.
- Verified functionality and throughput using FIO benchmarks across multiple configurations (single disk, RAID, XFS).
- Measured real-world database performance with PostgreSQL (pgbench) workloads.

Mar 2024 - Dec 2025

Selected Publications [Full list on Google Scholar]

- **BoostX™-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration** [URL]
The 53rd International Symposium on Computer Architecture (ISCA) Industry Track, 2026
Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, **Youngwoo Jeong**, Dongu Kim, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim
- **SEAM: A synergetic energy-efficient approximate multiplier for application demanding substantial computational resources** [URL]
Integration. vol.101, 2025
Youngwoo Jeong, Joungmin Park, Raehyeong Kim, Seung Eun Lee
- **Robot-Specific Processor for Autonomous Driving**
1st Workshop on Robotics Acceleration with Computing Hardware (RoboARCH) (Co-located with MICRO), Chicago, USA, Oct., 2022
Youngwoo Jeong, Kwang Hyun Go, Soohee Kim, Seung Eun Lee
- **An Architecture for Resilient Federated Learning through Parallel Recognition (Poster Session)** [URL]
The 31st International Conference on Parallel Architectures and Compilation Techniques (PACT), Chicago, USA, Oct., 2022
Jeongeun Kim, **Youngwoo Jeong**, Suyeon Jang, Seung Eun Lee
- **Photoplethysmography-Based Distance Estimation for True Wireless Stereo** [URL]
Micromachines. vol.14, no.2, 2023
Youngwoo Jeong, Joungmin Park, Sun Beom Kwon, Seung Eun Lee

Patents

Federated Learning Method and System Using Shared Learning Data

SEUNG EUN LEE, JEONGEUN KIM, YOUNGWOONG JEONG

patent application

United States

December 2023

Method and System for Determining Final Result Using Federated Learning

SEUNG EUN LEE, JEONGEUN KIM, YOUNGWOONG JEONG

patent application

United States

December 2023

Research Project

Development for Processing Software on AI Semiconductor Devices

MINISTRY OF SCIENCE AND ICT

- Analyzed various AI models to standardize the input for AI systems.
- Proposed an architecture for a hardware scheduler optimized for multi-AI core architecture.

South Korea

2024 - 2022

Development of Proximity/Healthcare Convergence Sensor SoC for TWS

MINISTRY OF TRADE, INDUSTRY AND ENERGY

- Designed a test environment for photoplethysmography sensors to evaluate their performance.
- Proposed a waveform adjustment filter and AI-based distance estimation algorithm to enhance signal processing accuracy.

South Korea

2023 - 2021

Embedded AI Module Based on Neuromorphic Computing

MINISTRY OF TRADE, INDUSTRY AND ENERGY

- Designed various applications utilizing multiple embedded AI modules.
- Developed a testbed for evaluating multi-AI core controllers.
- Proposed methodologies to enhance accuracy in federated learning with multi-AI core systems.

South Korea

2021 - 2020

Honors & Awards

School of Engineering Ph.D. Fellowship

NEW YORK UNIVERSITY (NYU)

New York, USA

August 2026

Excellent Thesis Award

SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

Seoul, South Korea

February 2024

- Topic:** Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller

Corporate (LX Semicon) Special Award

KOREA SEMICONDUCTOR INDUSTRY ASSOCIATION

Seoul, South Korea

October 2022

- Topic:** AI Processor employing Stochastic Computing for Embedded Systems

Department Chair Award

SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

Seoul, South Korea

February 2022

- Topic:** Design of an Autonomous Indoor Robot for Robot-on-Chip

Corporate (Silicon Mitus) Special Award

KOREA SEMICONDUCTOR INDUSTRY ASSOCIATION

Seoul, South Korea

November 2021

- Topic:** In-Vehicle Network Processor based on LIN and CAN-FD Controller

Teaching Experience

Advanced AI Processor, Computer Architecture

SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

Teaching Assistant

Seoul, South Korea

Fall 2022

Digital System Design, Resilient Processor Design

SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

Teaching Assistant

Seoul, South Korea

Spring 2022

Skills

Hardware Description Languages

Verilog, SystemVerilog

High-Level Computer Languages

SystemC, C, C++, Python, Matlab

Design and Implementation Tools

Catapult HLS, Design Compiler, IC Compiler II, Quartus II, Vivado

Verification and Analysis Tools

Verdi, VCS, ModelSim, PSpice, PrimeTime, Formality, StarRC

Benchmark Tools

Flexible I/O (FIO), PGbench

Design of Robot-Specific Processor for Autonomous Driving

- Technology: Samsung 28nm RFCMOS
- Designer: **Youngwoo Jeong**, Yue Ri Jeong, Hyun Woo Oh, Kwang Hyun Go
- Gate Counts: 1062K @ 50MHz
- Memory: Code region (16KB), Data region (128KB)
- Date: 2022. 07. 18

